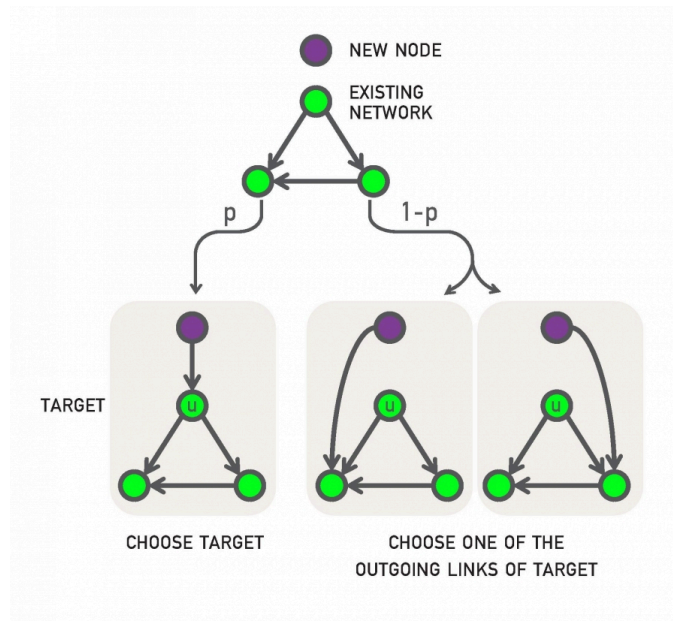


L12: Link-Copy Model



The PA model is simple (it has only one parameter) and it can generate a power-law degree distribution. However, the exponent of that distribution is fixed (equal to 3) and so the PA model does not give us the flexibility to adjust the exponent of the degree distribution to the same value that a given network has.

Let us now study another simple model that can also generate power-law networks but with any exponent we want. Additionally, this model can be used to create directed or undirected networks (we present it here in the context of directed networks).

The model is probabilistic and it also generates a growing network (one new node at a time) – as in the case of PA. Specifically, every time we add a new node v , it connects (with an outgoing edge) to an existing node as follows:

- With probability p , v connects to a randomly chosen existing node u .
- With probability $q=1-p$, v connects to a random node w that u connects to. In other words, v “copies” an outgoing edge of u (if u does not have any outgoing edges, v chooses another node. If no node has an outgoing edge, v connects to a random node).

The previous model is referred to as “link-copy” process. An equivalent way to describe the model is that, with probability $q=1-p$, node v selects a random edge in the network, say from a node u to a node w , and v connects to w .

The previous process can be repeated for $m>1$ new outgoing edges of node v . For simplicity, let us analyze the model for $m=1$ new edge.

Let us denote as $x_j(t)$ the in-degree of node j at time t . How does $x_j(t)$ increase with time?

There are two cases:

- node- j is selected randomly (with probability p) from a new node and it gains one incoming edge. At time t , there are $N=t$ nodes (recall that we add one node at each time unit), and so the probability that a node- j will get a new incoming edge at time t in this manner is p/t .
- an incoming edge of node- j is selected with probability q – and node- j gains again one incoming edge. The probability of that happening at time t is $\frac{q x_j(t)}{t}$ because node- j has $x_j(t)$ incoming edges at that time.

The following analysis will rely on the same assumptions we used in the PA model, i.e., continuous-time approximation and deterministic degree dynamics.

So, we can write a (deterministic) differential equation that expresses the rate at which node- j gains edges:

$$\frac{dx_j}{dt} = \frac{p}{t} + \frac{q x_j}{t} = \frac{p+q x_j}{t}$$

The initial condition for each node- j is that $x_j(t_j) = 0$, where t_j is the time that j was born.

If we index the nodes based on their time of birth (so that node-1 is born at time $t=1$, node-2 is born at time $t=2$, etc), we can write that $x_j(j) = 0$ for all j .

The previous differential equation is easy to solve if we rewrite it as

$$\frac{dx_j}{p+q x_j} = \frac{dt}{t}$$

and integrate both sides, we get:

$$\frac{1}{q} \ln(p + q x_j) = \ln t + c$$

where c is a constant. Exponentiating both sides of the equation, we get

$$p + q x_j = e^{c q t^q}$$

and so: $x_j(t) = \frac{1}{q}(e^{c q t^q} - p)$. The initial condition $x_j(j) = 0$ gives us that $e^{c q} = p/j^q$. So the solution for the in-degree dynamics is:

$$x_j(t) = \frac{p}{q} \left[\left(\frac{t}{j} \right)^q - 1 \right]$$

Note that the in-degree increases with time as a power-law with exponent q – contrast that with the PA model in which this exponent is fixed to $1/2$.

But what is the in-degree distribution for this model? Let us calculate first the complementary cumulative degree distribution $P[X_j > k]$, i.e., for a given in-degree k and a time t , what is the fraction of the t nodes in the network that have $x_j(t) > k$? In other words, we are asking what is the maximum value of j for which

$$x_j(t) = \frac{p}{q} \left[\left(\frac{t}{j} \right)^q - 1 \right] > k$$

which is equivalent to

$$j < t \left(\frac{q}{p} k + 1 \right)^{-1/q}$$

So, the fraction of nodes with degree larger than k is:

$$P[X_j > k] = \left(\frac{q}{p} k + 1 \right)^{-1/q}$$

To find the probability that a node has in-degree equal to k , we can differentiate the previous expression with respect to k :

$$p_k = P[X_j = k] = -\frac{dP[X_j > k]}{dk} = \frac{1}{p} \left(\frac{q}{p} k + 1 \right)^{-(1/q+1)}$$

This shows that the link-copy model creates a power-law degree distribution with exponent:

$$\frac{1}{q} + 1 = \frac{2-p}{1-p}$$

Note that when p approaches 1, the new node connects to random existing nodes and so there is no bias to connect to nodes with higher in-degree. In that case, the previous exponent diverges and the in-degree distribution decays exponentially fast with k .

As p approaches 0, on the other hand, the power-law exponent approaches 2, which means that both the first and second moment of the degree distribution diverge. The network acquires a *hub-and-spoke* topology in that case, with every new node connecting to the first node.

For intermediate value of p , we can get any desired exponent larger than 2. For example, for $p=1/2$, we get that the exponent is equal to 3, as in the PA model. This does not mean, however, that the networks generated by the PA model are identical with

the networks generated by the link-copy model for $p=1/2$.

The link-copy model has been used to explain the emergence of power-law networks in directed networks such as the Web, citation networks or gene regulatory networks. In all such networks there are “copy-edge” mechanisms.

For example, when someone creates a new Web page, it is often the case that he/she copies links of other relevant web pages.

Similarly, when writing the bibliography of a research article, authors are often “tempted” to copy references of other relevant articles (hopefully after they have read them!).

And in biology, the process of gene duplication creates multiple copies of the same gene. Those genes have the same promoters, and so, at least initially, they can be regulated by the same transcription factors.

Over time, mutations may change the promoter of a gene, creating regulatory differences in the incoming edges of different gene copies.

Food For Thought

Compare the network topology of the PA model with the undirected version of a Link-Copy network when $p=1/2$. Both models generate the same power-law degree distribution in that case. How are the two networks different however?